DDA 3020 HW 1

Q1 (a). Denote $\tilde{w} = (w, b)^T \in R^{(d+1)\times m}$, $X = \begin{pmatrix} 1 & \tilde{x}_1^T \\ \vdots & \vdots \\ 1 & \tilde{x}_N^T \end{pmatrix} \in R^{N\times(d+1)}$

$J(w,b) = J(\tilde{w}) = tr[(Y - X\tilde{w})^T A (Y - X\tilde{w})]$

$$\frac{dJ}{d\tilde{w}} = \frac{d}{d\tilde{w}}(Y^TAY - 2Y^TAXW + W^TX^TAXW)$$

$$= -2X^TAY + 2X^TAXW$$

when $\frac{dJ}{d\tilde{w}} = 0$

$$\Rightarrow \tilde{w} = (X^TAX)^{-1}X^TAY$$

where $\tilde{W} = (w, b)^T$

(b). $\frac{dJ}{d\tilde{w}} = -2X^TA(Y - X\tilde{w})$

$$\Rightarrow \frac{dJ_i}{d\tilde{w}} = -2\alpha_i(y_i - \tilde{w}^T\tilde{x}_i)\tilde{x}_i^T$$

$$\Leftrightarrow \frac{\partial J}{\partial W} = -2\sum_{i=1}^{N}\alpha_i(y_i - (Wx_i + b))x_i^T$$

$$\frac{\partial J}{\partial b} = -2\sum_{i=1}^{N}\alpha_i(y_i - Wx_i - b)$$

$\Rightarrow$ GDM:

$$W \leftarrow W + 2\eta\sum_{i=1}^{N}\alpha_i(y_i - (Wx_i + b))x_i^T$$

$$b \leftarrow b + 2\eta\sum_{i=1}^{N}\alpha_i(y_i - (Wx_i + b))$$

where η is the learning rate

Q2 (a). ① $\nabla f_0 = \begin{pmatrix} 2(x_1 - 2) \\ 2(x_2 - 1) \end{pmatrix}$ $\nabla^2 f_0 = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$

$\nabla^2 f_0$ is Positive Definite

$\Rightarrow f_0$ is a convex function

②. Constraits are all linear

i.e. $x_1 + x_2 - 1 \le 0$, $-x_1 \le 0$, $-x_2 \le 0$

$\Rightarrow$ x $\in$ convex set

$\Rightarrow$ from ①, ②, it's a convex opt problem

$$L(x, \mu) = (x_1 - 2)^2 + (x_2 - 1)^2 + u_1(x_1 + x_2 - 1) - \mu_2 x_1 - \mu_3 x_2$$

KKT conditions

$$\begin{cases} \nabla_x L = \begin{pmatrix} 2(x_1 - 2) + u_1 \\ 2(x_2 - 1) + u_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} & ① \\ x_1 + x_2 \le 1, \ x_2 \ge 0, \ x_3 \ge 0 & ② \\ u_1, u_2, u_3 \ge 0 & ③ \\ u_1(x_1 + x_2 - 1) = 0, \ u_2 x_1 = u_3 x_2 = 0 & ④ \end{cases}$$

(b). From ①, $\begin{cases} x_1 = 2 - \frac{u_1}{2} \\ x_2 = 1 - \frac{u_1}{2} \end{cases}$

1° suppose $u_1 = 0$

$\Rightarrow x_1 = 2, x_2 = 1$

$x_1 + x_2 = 3 > 1$ Contradiction!

2°. suppose $x_1 + x_2 - 1 = 0$

$\Rightarrow 2 - \mu_1 = 0$

$\Rightarrow u_1 = 1$

$x_1 = 1, x_2 = 0$, which satisfies all KKT conditions

$\Rightarrow x^* = (1, 0)$

(c)(1) $LHS = D_{KL}(P\|Q) = \sum_x P(x)\log\frac{P(x)}{Q(x)}$

$RHS = D_{KL}(Q\|P) = \sum_x Q(x)\log\frac{Q(x)}{P(x)}$

to prove $LHS - RHS = \sum_x (P(x) + Q(x))\log\frac{P(x)}{Q(x)} \ne 0$

$$\Leftrightarrow \prod_x \left(\frac{P(x)}{Q(x)}\right)^{P(x)+Q(x)} \ne 1$$

Since the geometric mean of $\frac{P(x)}{Q(x)}$ might not be 1 (e.g. $P(x) = \begin{cases} \frac{1}{3}, & x=1 \\ \frac{2}{3}, & x=2 \\ 0, & \text{otherwise} \end{cases}$

$Q(x) = \begin{cases} \frac{1}{2}, & x=1 \\ \frac{1}{2}, & x=2 \\ 0, & \text{otherwise} \end{cases}$

$\Rightarrow LHS - RHS = \frac{5}{6}\log\frac{2}{3} + \frac{7}{6}\log\frac{4}{3} \approx -0.002 < 0$)

then the inequality holds

(2). $P(x) = \begin{cases} \frac{1}{3}, & x=1 \\ \frac{2}{3}, & x=2 \\ 0, & \text{otherwise} \end{cases}$

$Q(x) = \begin{cases} \frac{1}{2}, & x=1 \\ \frac{1}{2}, & x=2 \\ 0, & \text{otherwise} \end{cases}$

$LHS - RHS = \frac{5}{6}\log\frac{2}{3} + \frac{7}{6}\log\frac{4}{3} \approx -0.002 \ne 0$)

Q3 (a). $\sigma(z) = \frac{1}{1+e^{-z}}$

$$\sigma'(z) = -(1+e^{-z})^{-2}(e^{-z}\cdot -1)$$

$$= \frac{e^{-z}}{(1+e^{-z})^2}$$

$$= \sigma(z)\cdot\frac{1+e^{-z}-1}{1+e^{-z}}$$

$$= \sigma(z)(1 - \sigma(z))$$

(b). $\frac{\partial J_i}{\partial(w)} = \frac{\partial J_i}{\partial h_i} \cdot \frac{\partial h_i}{\partial(w^T x_i)} \cdot \frac{\partial(w^T x_i)}{\partial w}$

$= -\left(\frac{y_i}{h_i} - \frac{1-y_i}{1-h_i}\right) \cdot h_i(1-h_i) \cdot x_i$

$= (h_i - y_i) \cdot x_i$

$\Leftrightarrow \frac{\partial J}{\partial(w)} = \sum_{i=1}^{n} (h_i - y_i) x_i$

$= X^T(h-y)$, where $h = \sigma(Xw)$

(c). $H = \nabla_w(X^T(h-y))$

$= \frac{\partial(X^T(h-y))}{\partial h} \cdot \frac{\partial h}{\partial w}$

$= X^T \cdot \begin{pmatrix} h_1(1-h_1)x_1^T \\ \vdots \\ h_n(1-h_n)x_n^T \end{pmatrix}$

$= X^T S X$, where $S = \mathrm{diag}(h_i(1-h_i))$, $i \in [1, n]$

$\forall v \in R^d$

$v^T H v = (Xv)^T S (Xv) = \sum_{i=1}^{n} S_i (Xv)_i^2$

Since $S_i = h_i(1-h_i) \geq 0$, $h_i \in (0,1)$

$(Xv)_i^2 \geq 0$

$\Rightarrow v^T H v > 0$

$\Rightarrow H$ is PSD

$\Rightarrow J(w)$ is convex

(d). $J_{reg}(w) = J(w) + \frac{\lambda}{2} w^T w$

$\nabla_w J_{reg} = X^T(h-y) + \lambda w$

GDM: $w^{(t+1)} = w^{(t)} - \alpha\left(X^T(h^{(t)}-y) + \lambda w^{(t)}\right)$

where α is the learning rate

(e). $L = -\sum_{i=1}^{n} \sum_{k=1}^{K} y_{ik} \log P_{ik}$

$= -\sum_{i=1}^{n} \sum_{k=1}^{K} y_{ik}\left(w_k^T x_i - \log \sum_{j=1}^{k} e^{w_j^T x_i}\right)$

Since $\frac{\partial\, -\sum_{i=1}^{n} \sum_{k=1}^{K} y_{ik}(w_k^T x_i)}{\partial w_k} = -\sum_{i=1}^{n} y_{ik} x_i$ ①

$\frac{\partial \sum_{i=1}^{n} \sum_{k=1}^{K} y_{ik} \log \sum_{j=1}^{k} e^{w_j^T x_i}}{\partial w_k}$

$= \sum_{i=1}^{n} \frac{\partial}{\partial w_k} \log \sum_{j=1}^{k} e^{w_j^T x_i}$

$= \sum_{i=1}^{n} \frac{1}{\sum_{j=1}^{k} e^{w_j^T x_i}} \cdot \frac{\partial}{\partial w_k}\left(\sum_{j=1}^{k} e^{w_j^T x_i}\right) = \sum_{i=1}^{n} \frac{e^{w_k^T x_i}}{\sum_{j=1}^{k} e^{w_j^T x_i}} x_i = \sum_{i=1}^{n} P_{ik} x_i$ ②

① + ②, we have $\nabla_{w_k} L = \sum_{i=1}^{n} (P_{ik} - y_{ik}) x_i$

$= X^T(P_{ik} - y_{ik})$

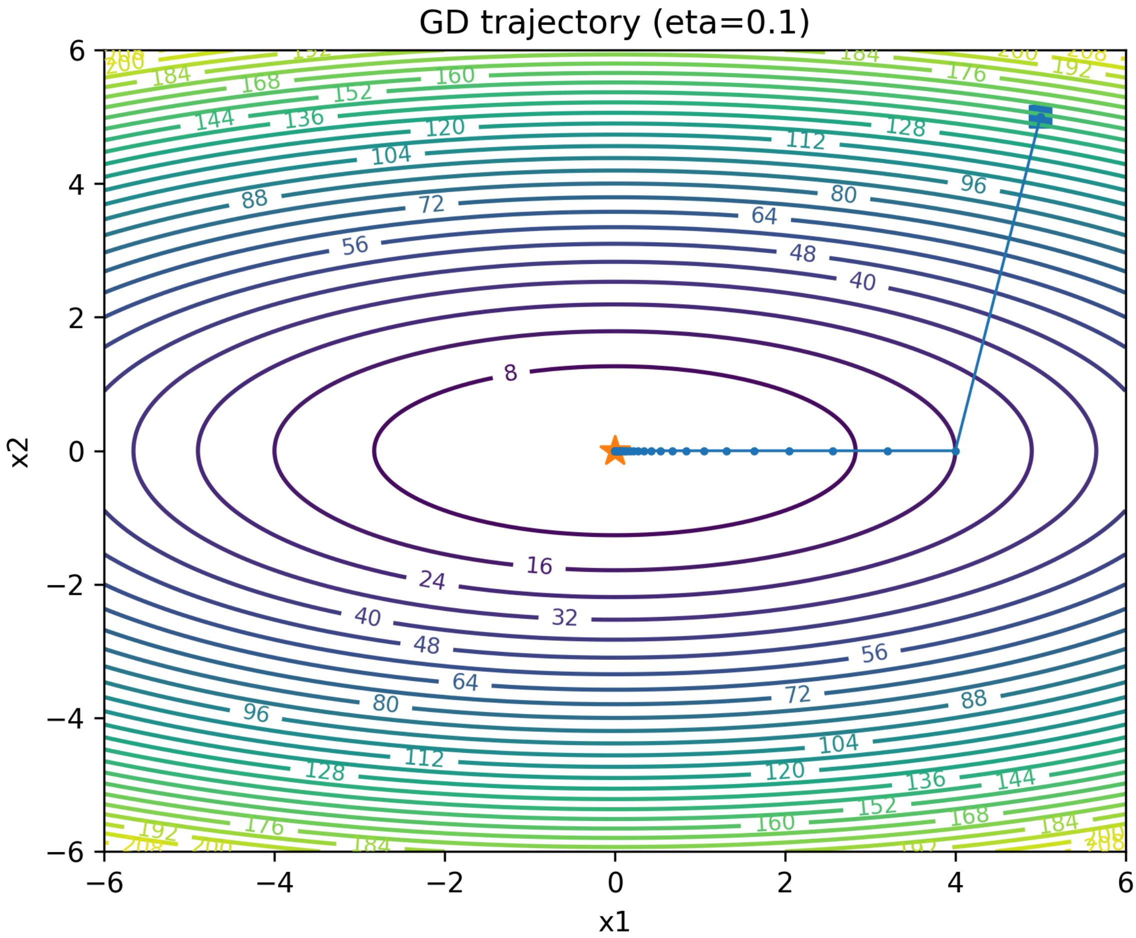
DDA3020 HW1

2.1. Gradient Descent from Scratch

(a)

x1	$0.0000713624 \approx 0$
x2	$0.0000000000 \approx 0$
f(x)	$5.092589940836e-09 \approx 0$

(b)



(c)

eta	iterations
0.01	converged in 798 iterations
0.1	converged in 73 iterations
1.1	diverged by iter 10000

- Reason for divergence for eta = 1.1:

$$x_2^{(t+1)} = (1 - 10\eta)x_2^{(t)}$$

  - For eta = 1.1, we have $x_2^{(t+1)} = (1 - 11)x_2^{(t)} = -10x_2^{(t)}$. This means that the value of x_2 will grow exponentially in magnitude, leading to divergence.

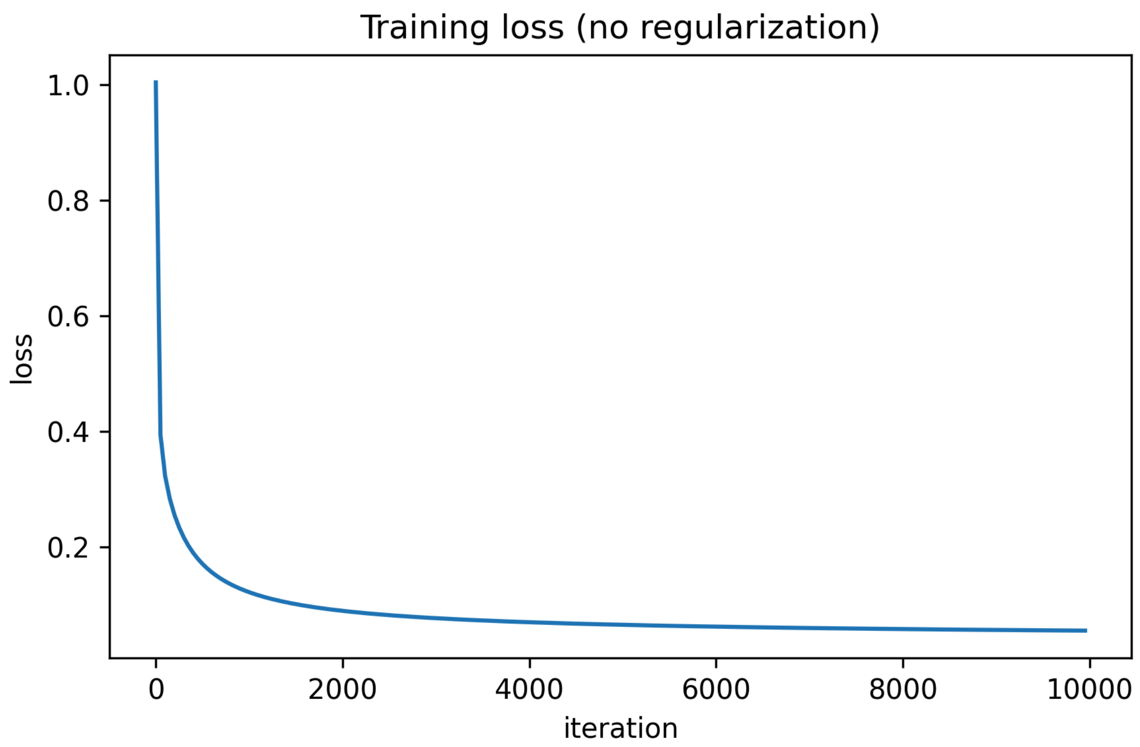
2.2. Logistic Regression with Regularization

> ⚠ **Warning**
>
> - *Seed = 1*
> - *Hyperparameters: lr (eta)=0.1, epochs=10000*

> **Important**
>
> Aligned with lecture notes, cost functions for L1 and L2 regularization are defined as follows:
>
> - $$L2 : \bar{J}(w) = J(w) + \frac{\lambda}{2m}\sum_{j=1}^{d} w_j^2$$
> - $$L1 : \bar{J}(w) = J(w) + \frac{\lambda}{m}\sum_{j=1}^{d} |w_j|$$

(a)

Train acc = 0.9750, Test acc = 1.0000



(b) L2

λ	Train acc	Test acc	n(zero weights)
0.01	*0.9750*	*1.0000*	0
0.1	*0.9750*	*1.0000*	0
1.0	*0.9500*	*0.9667*	0

(c) L1

λ	Train acc	Test acc	n(zero weights)
0.01	*0.9750*	*1.0000*	0
0.1	*0.9750*	*1.0000*	0
1.0	*0.9667*	*1.0000*	4

(d)

Comparison:

- Model Performance:
    - Both L1 and L2 regularization with λ values of 0.01 and 0.1 achieved perfect test accuracy (1.0000) and high train accuracy (0.9750).
    - However, at $\lambda = 1.0$, L2 regularization resulted in a decrease in both train and test accuracy (0.9500 and 0.9667), while L1 regularization maintained a higher test accuracy (1.0000) despite a slight drop in train accuracy (0.9667).
- Sparsity:
    - L1 regularization with $\lambda = 1.0$ resulted in 4 out of 12 weights being near-zero, indicating that L1 regularization promotes sparsity in the learned weights.
    - In contrast, L2 regularization did not produce any near-zero weights across all λ values, suggesting that L2 regularization does not encourage sparsity and rather tends to shrink weights.